Juan Antonio Monleón de la Lluvia

Nuclear Engineer · PhD Candidate in Nuclear Data & Neutronics

CONTACT

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EDUCATION

PhD in Nuclear Data & Neutronics
Université Paris-Saclay · ED PHENIICS — 2023-2026

Double Master in Industrial Engineering and Nuclear Science and Technology
Polytechnic University of Madrid — 2021-2023

Master in Management and Direction of SMEs
EFEM Business School — 2024-2025

Bachelor in Industrial Technology Engineering
Polytechnic University of Madrid — 2017-2021

TECHNICAL SKILLS

Nuclear Codes: MCNP, Serpent, OpenMC, NJOY, ADVANTG

Programming: Python, Git, Claude Code

Documentation: LaTeX, Microsoft 365

ABOUT ME

Nuclear engineer working on nuclear data and on the sensitivity and uncertainty analysis of neutronic simulations, from the evaluation of covariances to their propagation in shielding and reactor-dosimetry calculations. Proactive and quick to adapt to new codes, tools and problem domains.

RESEARCH EXPERIENCE

PhD in Nuclear Data & Neutronics

ASNR, France · Université Paris-Saclay — Nov 2023 - Present

Thesis: **"Nuclear-Data Uncertainty Quantification for PWR Vessel Aging with a ^{56}Fe Elastic-Scattering Evaluation"** *(defence scheduled 1 October 2026)*

- Sensitivity analysis and Monte Carlo sampling to propagate nuclear-data uncertainties; re-evaluated ^{56}Fe elastic angular distributions and covariances from EXFOR data
- Applied variance-reduction techniques (ADVANTG weight windows) to improve efficiency in deep-penetration shielding calculations
- Supervised a Master's student (Mar–Sep 2025) on nuclear-data uncertainty quantification in PWR source-term calculations

Master's Thesis Research (MII)

UPM, Spain — Jun 2023 - Oct 2023

- Developed **"Enhancing the EXFOR nuclear data library with Machine Learning Techniques"**
- Implemented outlier detection methods for proton-induced reactions

Research Internship — Master's Thesis (MUCTN)

SCK CEN, Belgium — Sep 2022 - Jun 2023

- Completed Master's thesis titled **"Reducing Spectrum-Driven Uncertainties with Variance Reduction Techniques"**
- Conducted neutronic and deep shielding analyses using MCNP and ADVANTG for the MYRRHA Project

Master Internship

ENUSA Advanced Industries, Spain — Oct 2021 - Apr 2022

- Designed and developed a Python program for the analysis and inspection of fuel assemblies and spent fuel

LANGUAGES

Spanish Native language

English C2 Level

French C1 Level

AWARDS

Best Academic Report for Nuclear Master
Polytechnic University of Madrid (2022-2023)

Best Master Thesis Award
SCK CEN, Belgium (2023)

Best Poster Award
Journées des Thèses of IRSN (2024) - **View Poster**

COMMUNITY

JEFF Community
Active member · Joint Evaluated Fission and Fusion File (OECD/NEA)

INDEN Community
Member · International Nuclear Data Evaluation Network (IAEA)

SINBAD Community
Member · Shielding Integral Benchmark Archive & Database (OECD/NEA)

REFERENCES

Dr. Mariya Brovchenko
ASNR, France
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Dr. Dimitri Rochman
Nagra, Switzerland
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Prof. Óscar Cabellos
Universidad Politécnica de Madrid, Spain
oscar.cabellos@upm.es

PUBLICATIONS

- **J. A. Monleon de la Lluvia, M. Brovchenko, D. Rochman, E. Dumonteil**, **“Towards Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations”**, *Nuclear Science and Engineering*, 2025. DOI: 10.1080/00295639.2025.2510048 · **pre-print** — 2025

CONFERENCES & PRESENTATIONS

FIRST AUTHOR

- **“Evaluation of elastic angular distribution covariances for ^{56}Fe from EXFOR experimental data”** — WONDER 2026, Aix-en-Provence *(upcoming)* — Jun 2026
- **“Impact of Elastic Scattering Angular Distribution Covariances on Fast Neutron Fluence; Application to ^{56}Fe”** — PHYSOR 2026 — Apr 2026
- **“Nuclear Data Uncertainty Quantification Using the VENUS-3 Benchmark Experiment”** — ND 2025, Madrid — Jun 2025
- **“Sensitivity Analysis and Uncertainty Quantification in PWR Irradiation Ageing-like problems”** — ANS Winter Conference RPSD, Orlando — Nov 2024

CO-AUTHORED

- **“Sensitivity Coefficients of a Single Response Function in Fixed-Source Calculations Using SERPENT2 and MCNP6”** — PHYSOR 2026 — Apr 2026
- **“Propagation of Angular Distribution Uncertainties in Vessel Fluence Calculations”** — PHYSOR 2026 — Apr 2026

SOFTWARE & TOOLS

KIKA — Desktop Workspace for Nuclear Data

Author & lead developer — 2025 - Present

- Opens ENDF, ACE, EXFOR and SDF files, drives a local NJOY installation, reads MCNP and Serpent runs, and carries the results into nuclear-data sampling, sensitivity and uncertainty analysis, all in a single desktop application
- Also author of **kika-nd**, the open-source Python library (PyPI) I wrote to provide the computational core behind the application

TRAINING

- OpenMC Course — NEA, Paris — Dec 2025
- NJOY Course — NEA, Paris — Dec 2024
- Sensitivity Analysis Summer School (SAMO) — UNIPR, Parma — Jun 2024
- Serpent Bootcamp — NEA, Paris — Nov 2023
- Course in "Nuclear Data for Energy and Non-Energy Applications" — GREAT PIONEER — Dec 2022